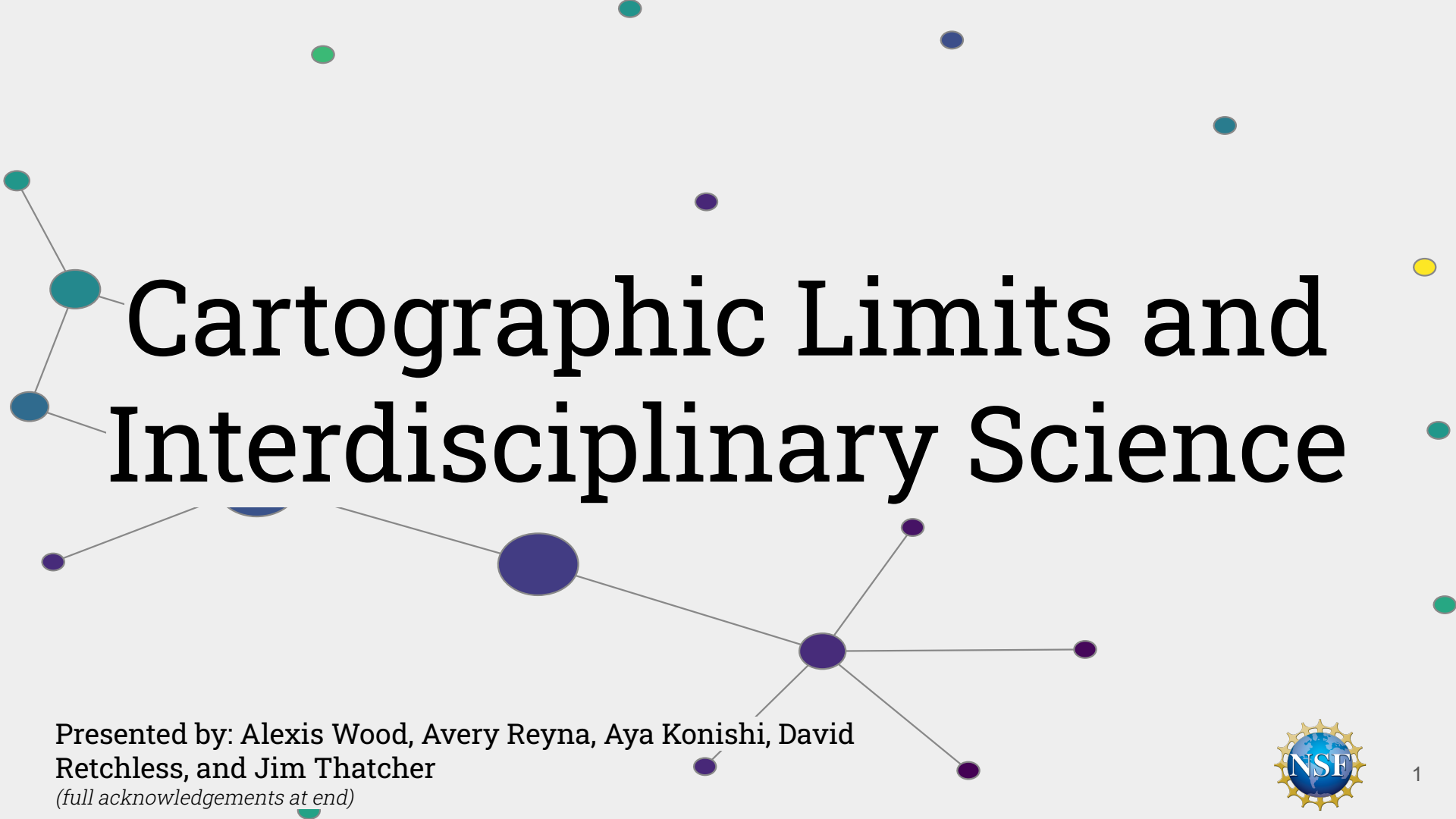




Cartographic Limits and Interdisciplinary Science

Presented by: Alexis Wood, Avery Reyna, Aya Konishi, David
Retchless, and Jim Thatcher
(full acknowledgements at end)



Talk Outline

- The project
 - A very brief demonstration of gerrymandering
 - Finding and defining Communities of Interest for Electoral Districting
- Cartographies role in interdisciplinary work
 - A tool of communication?
 - ... or something more?
- Seeing lines, pushing limits
- Translational science beyond the cartesian grid

The Project:

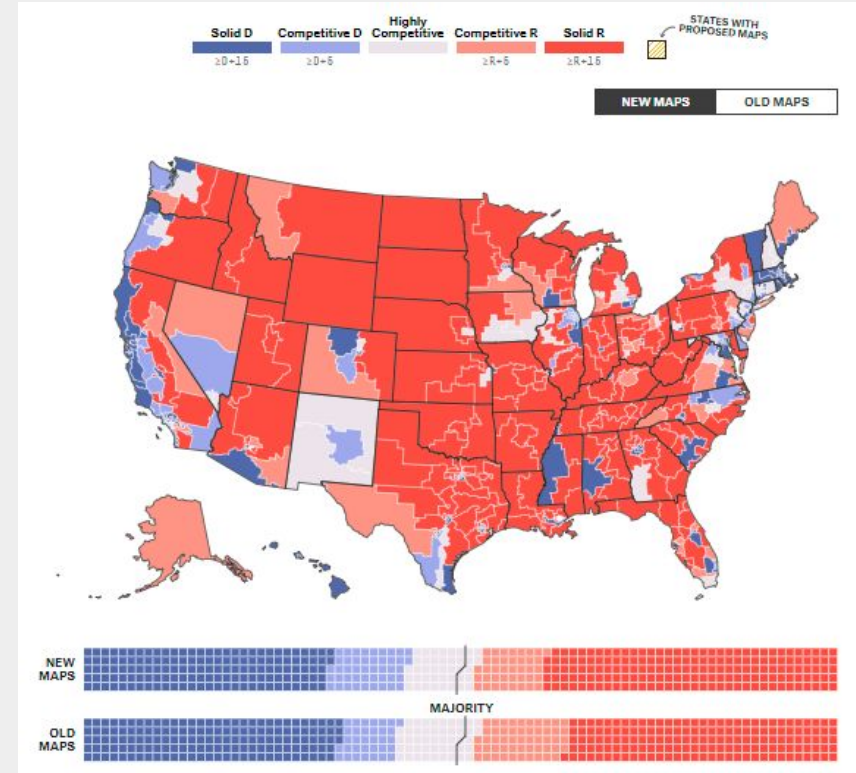
Spatial Models and Electoral Districting

- National Science Foundation Research Experience for Undergraduates Site (Award #1851696)
- Ran from 2019-2022
- Interdisciplinary approaches to questions of electoral districting
 - Mathematics, computer science, data science, critical human geography, **cartography**, and GIS
- Today, discussion of communities of interest, cartography, and the limits thereof.

The Project:

What is Electoral Districting?

- Every 10 years the US conducts a census and redraws congressional districts
- Different actors draw lines in different states: legislatures, independent commissions, courts, etc.
- Many different laws govern the drawing of districts



The Project

All districts are gerrymandered

Maps are how we make claims upon the world

For good and ill, for justice and not; we use maps to define our worlds.

Where can you put in that new fence? What country do you live in? What's your tax rate?

How and what we map matters, and neither are given

There's no 'correct' district. They are socio-political constructions through which we organize our government and resources.

So the goal is something we **agree** is fair, not something that **is**.

Redistricting Criteria: Rules of the game

Equal Population

Racial Equity

Contiguity

Compactness

Partisan Bias

Protection of
Incumbents

Preservation of Local
Government

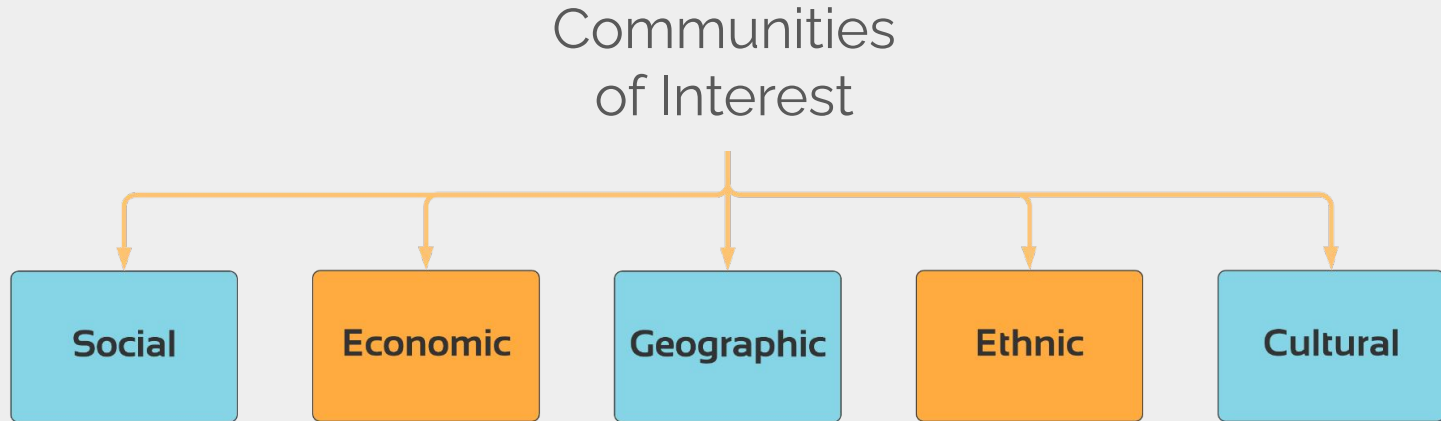
Preservations of
District Core

Preservation of
Communities of
Interest

*Laws governing redistricting are not consistent throughout all states

Communities of Interest

- Some, though not all, of the ways we might think of Communities of Interest:



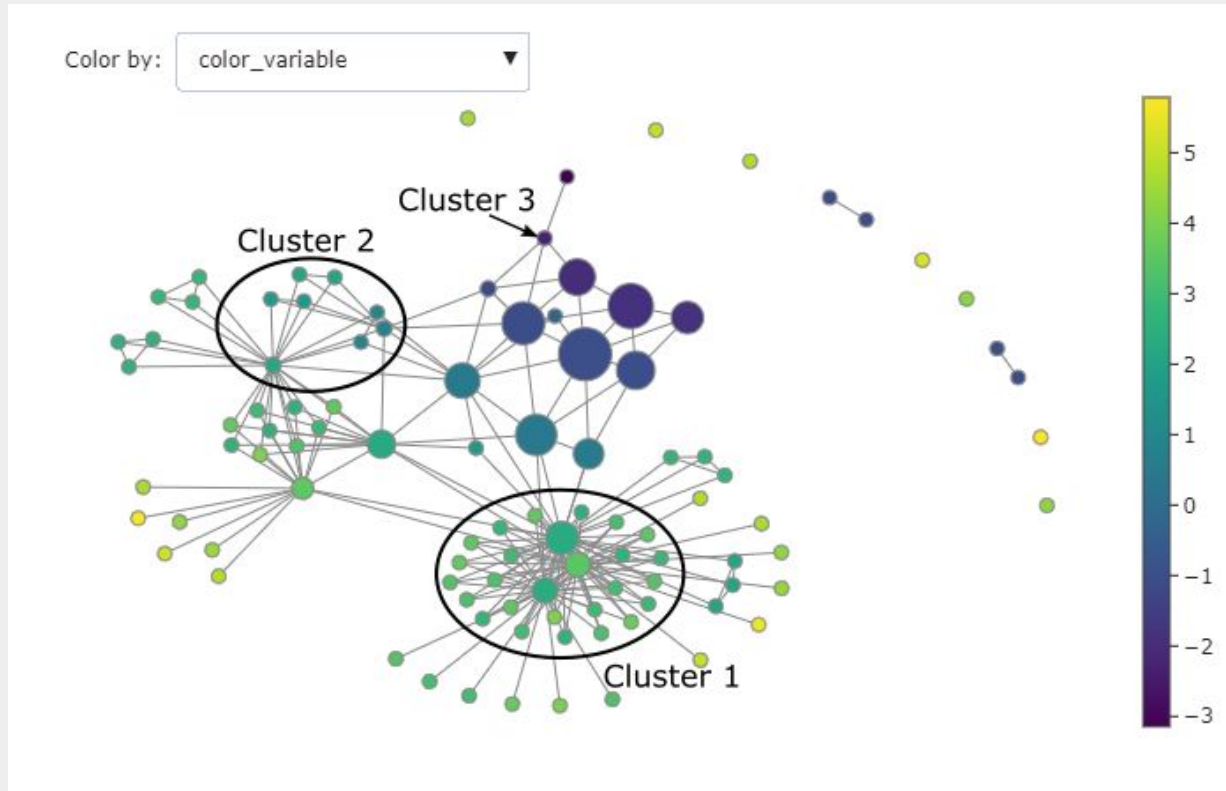
Communities of Interest



COI Mapped in Representable

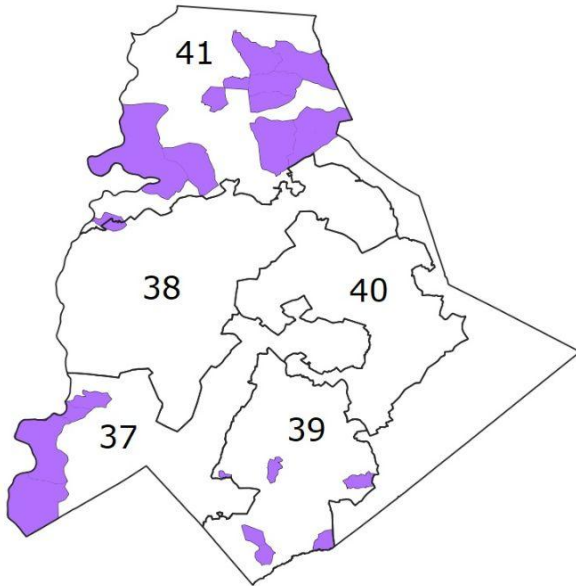
"Our family, both children and adults, attend many cultural events in EXETER, FARMERSVILLE, WOODLAKE and VISALIA, which are currently in a separate congressional district from where we live. We attend such events every week of the year. We NEVER attend such events in Bakersfield, and rarely (maybe once a year) in Porterville".

MAPPER and Communities of Interest

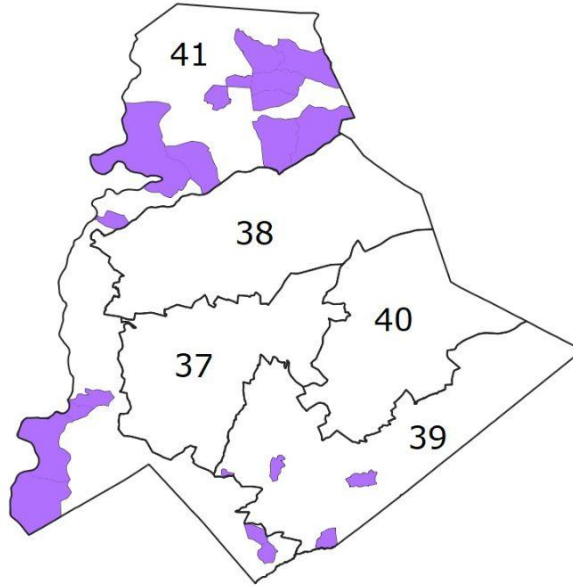


Cluster 3: 'Wealthy Homeowners with Children'

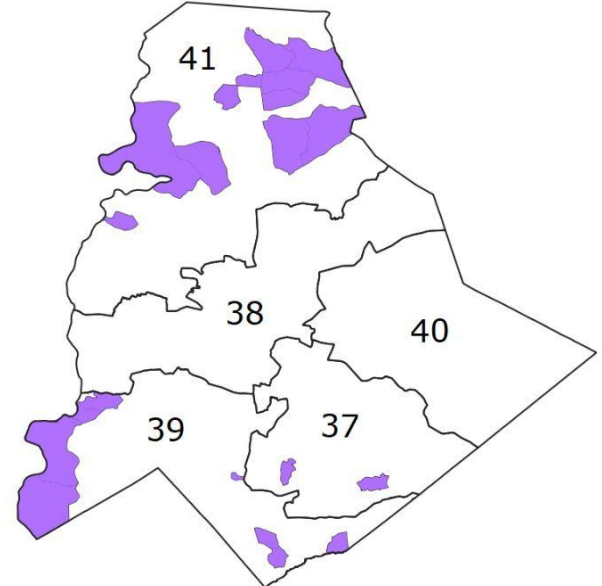
State Senate Plans



2011 Decennial Census based



2018 Court Ordered



2019 NC General Assembly

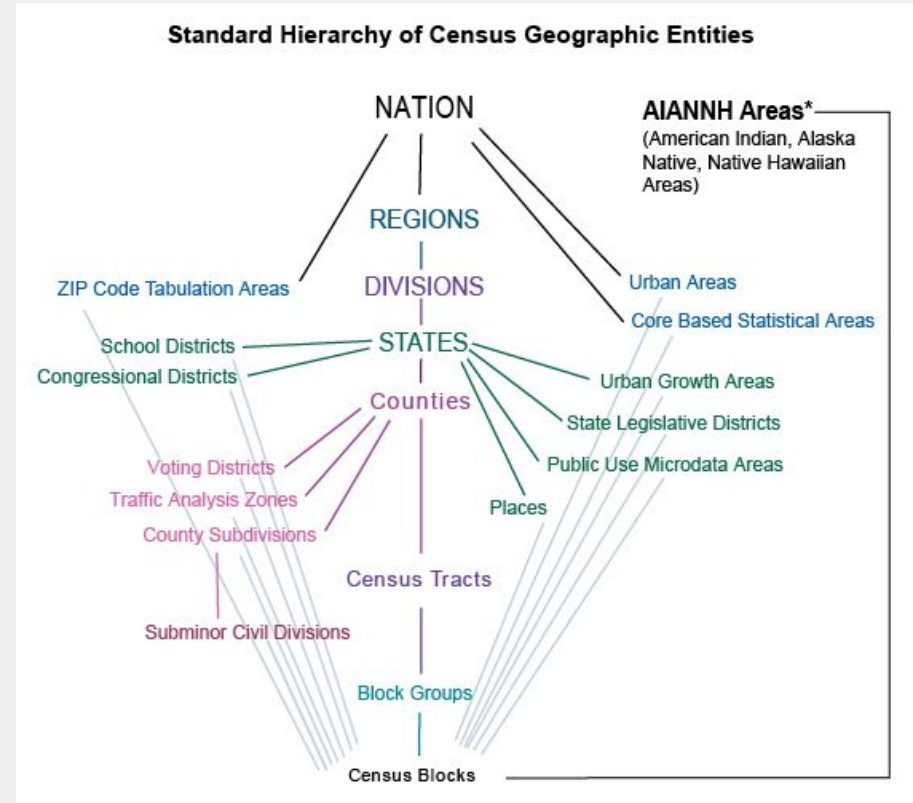
Tension

Communities of Interest

Ideas v Application

What have we actually accomplished and why?

The way we have approached incorporating Communities of Interest **has contributed to the problem it has meant to address.**



Cartography's Role in Interdisciplinary Work

- Often 'reduced' to two (important) roles:

- Data Preparation and correction
 - *Is this line in the correct place? What data format does this need to be in?*
- Final presentation and communication
 - *Make it look pretty!*

- **Both presented as perfunctory, straightforward**

- **Both deeply ontological**

'All ontological claims involve a certain epistemic violence - a 'closing off' - that is at once a necessary and problematic condition of knowledge' (Braun 1998, 717)

- Need pathways that recognize these limits
- And, to do that, we must push beyond the cartesian plane of data cleaning and clear communication